Lagoa Spring 2025

# EE 350

# Test 3

Closed Book, Closed Notes, 3 Sheets $8\frac{1}{2}$ in $\times$ 11 in

Write Clearly and Fully Justify Your Answers

**Question 1 (30%):** Consider the periodic **real** signal

$$f(t) = \sum_{n=-\infty}^{+\infty} D_n e^{j2nt}$$

where

$$D_0 = 1 \quad \text{and} \quad D_n = \frac{1}{n} e^{j(n^2+1/2)\pi} \text{ for } n \geq 1.$$

a) (10%) Determine the value of $D_n$ for $n \leq -1$.

b) (10%) Is the signal $f(t)$ even, odd or neither? Fully justify your answer

c) (10%) Now consider the block diagram below

$$f(t) \longrightarrow \boxed{H} \longrightarrow y(t)$$

where the system H has impulse response whose Fourier transform is

$$H(\omega) = \text{rect}(\omega/14)e^{-j3\omega}$$

Determine the power $P_y$ of the output $y(t)$.

**Question 2 (30%):** Here we consider the problem of sampling and recovering audio signals. The different parts consider different steps need to analyze when we can recover a signal after it has been sampled.

As part of this, define the periodic signal (with period $T_0$)

$$\delta_{T_0}(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT_0)$$

a) (10%) Compute the exponential Fourier series of $\delta_{T_0}(t)$ and use it to compute its Fourier transform. Compare the results with the one provided in the Fourier transform table.

b) (10%) Now consider an audio signal $f(t)$. Define the sampled signal $f_{\textbf{sampled}}(t)$ as the multiplication of the two signals

$$f_{\textbf{sampled}}(t) = f(t)\delta_{T_0}(t)$$

Compute the Fourier transform of $f_{\textbf{sampled}}(t)$ as a function of the Fourier transform of $f(t)$ (which we usually denote as $F(\omega)$).

c) (10%) Audio signals have a bandwidth of 20 kHz; i.e., we have

$$F(\omega) = 0 \text{ when } |\omega| \geq 4 \times 10^4\pi \text{ rad/s}$$

Let $f_{\textbf{sampled}}(t)$ be as in part (b). One can argue (from the properties of $\delta(t)$) that it has the form

$$f_{\textbf{sampled}}(t) = \sum_{n=-\infty}^{+\infty} f(nT_0)\delta(t - nT_0)$$

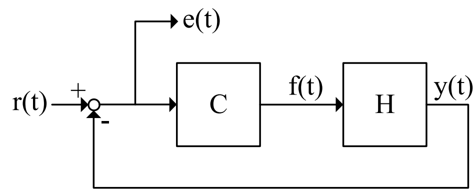
in other words it only contains information on the signal $f(t)$ at the sampling instants $0$, $\pm T_0$, $\pm 2T_0$, ... So, $f_{\textbf{sampled}}(t)$ can be thought as a mathematical representation of the sampling procedure used to produce digital signals that can be stored and used in digital devices.

Now, we want to be able to recuperate $f(t)$ from its samples $f_{\textbf{sampled}}(t)$. Using a reasoning based on Fourier transform, compute the all the values of $T_0$ that allow for the one to completely recover $f(t)$ and explain how we can recover $f(t)$ by filtering $f_{\textbf{sampled}}(t)$ with a suitable filter.

**Hint:** Sketch the Fourier transform of $f_{\textbf{sampled}}(t)$ as a function of $F(\omega)$ and use the fact that $F(\omega)$ is band limited at $4 \times 10^4\pi$ rad/s.

**Question 3 (30%):** In this problem, we aim at designing controllers that are able to follow constant reference signals with zero steady state error.

Consider the following closed loop system




where $H$ is a system described by the differential equation

$$\frac{dy}{dt} + 2y(t) = 3f(t)$$

and the controller $C$ has transfer function

$$C(\omega) = \frac{1}{j\omega + a}$$

where $a \geq 0$ is the controller parameter to be designed.

a) (10%) Compute the transfer function of the closed loop

$$H_{cl}(\omega) = \frac{E(\omega)}{R(\omega)}$$

and the differential equation that describes the relation between $r(t)$ and $e(t)$.

b) (10%) Compute the modes of the system as a function of $a$ and argue that the closed loop system is BIBO stable for all $a \geq 0$.

If you did not do part (a) use

$$H_{cl}(\omega) = \frac{(j\omega + a)(j\omega + 5)}{(j\omega + a)(j\omega + 5) + 4}$$

c) (10%) Now consider input

$$r(t) = \beta$$

where $\beta$ is a constant. Determine the value of $a$ leading to steady state value of $e(t)$

$$e_{ss}(t) = 0.$$

In other words, find the value of $a$ for which the steady state of the output $y(t)$ is equal to the input when the input is constant.

If you did not do part (a) use

$$H_{cl}(\omega) = \frac{(j\omega + a)(j\omega + 5)}{(j\omega + a)(j\omega + 5) + 4}$$

**Question 4 (10%):** Consider

$$\text{f(t)} \longrightarrow \boxed{\text{H}} \longrightarrow \text{y(t)}$$

and assume that

i) The signal $f(t)$ is periodic

ii) The transfer function of the system H satisfies $|H(\omega)| \leq 1$ for all $\omega$.

Is $P_y \leq P_f$? If YES, prove this. If NO, provide an example of a system H and a signal $f(t)$ where this does not hold.